

Chapter - 1+2

* Data Flow $\rightarrow$ simplex, half duplex, full duplex.

* Data communication components - Message, Sender, Receiver, Medium, Protocol

* Data Representation - text, numbers, images, audio, video.

* Network Type - PAN, LAN, MAN, WAN

$\downarrow$
Bluetooth

$\downarrow$
(50-300)m

$\downarrow$
Wimax

$\downarrow$
Internet

* Network Topology

Mesh

cable - $n(n-1)/2$

$\rightarrow$ No traffic load
 $\rightarrow$ Privacy & Security
 $\rightarrow$ High cost.

Star

cable - n

$\rightarrow$ Less than mesh
 $\rightarrow$ hub down so all down

Bus

Easy Installation

Ring

cable - n

$\rightarrow$ Privacy
 $\rightarrow$ Traffic

* Switch - Same network $\rightarrow$ no device connect $\rightarrow$

* Router - $\rightarrow$ different network connect $\rightarrow$

* Difference between switch & hub?

Ans. Hub carries information from $\rightarrow$ network

Switch carries information certain device (in network).

= De facto - Not approved by any organization - TCP/IP model.

De jure - Approved by $\odot$ organization - OSI model (ISO)

Address $\rightarrow$ Port address - 16 bit, $2^{16} \rightarrow$ S, D always same.
 $\rightarrow$ Logical/Ip address - 32 bit, 128 bit $\rightarrow$ S, D always same
 $\rightarrow$ MAC address - 48 bit $\rightarrow$ S, D differ in every hop

7 layer ad OSI model :

① Connection between two layers.

7 ● Application layer →

6 ● Presentation layer →

① Translation → *डाटा को convert*

② Compression → *डाटा को compress करे*

③ Encryption → *डाटा को encrypt करे*

5 ● Session layer → ① session maintain करे

→ ② Authentication.

③ Authorization

● Transport layer → ① Segmentation & re assembly

② Add port address & Sequence number

③ Connection Control.

④ Flow & Error Control.

⑤ Multiplexing

Port Address करे

sequence number.

(Segments)

3 ● Network layer → # Logical Address or Ip address

Routing

(Packet)

● DataLink layer → ① Framing

② Physical Addressing

③ Flow Control.

④ Access Control.

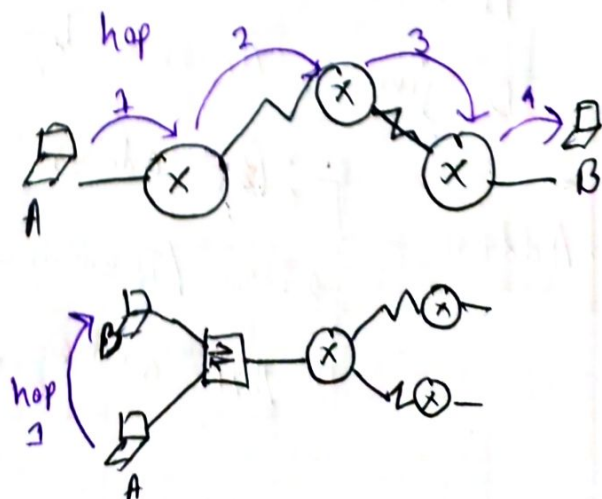
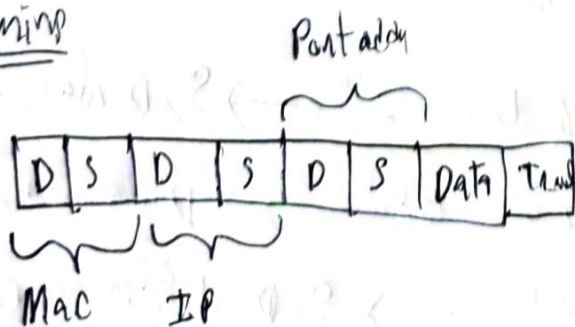
MAC address

(Frame)

● Physical layer → ① Data (in electric pulse) convert करे medium a *medium*

(Bit)

Addressing



Chapter - 3th

- Bandwidth $\rightarrow$ ① difference between highest frequency and lowest frequency.
② Range of frequency in a composite signal.
- Throughput $\rightarrow$ amount of data/bits that being transmitted through the link/media at a certain time.
- Latency $\rightarrow$ total time taken for a datagram to be sent.

1 Byte = 8 bits

Formula:

* Frequency = $1/\text{time period}$

* Bandwidth = Highest frequency - Lowest frequency

* No of levels = $2^{\text{no. of bits}}$

* Attenuation/Amplification = $10 \log_{10} \left(\frac{P_2}{P_1} \right)$

* SNR = $\frac{P_{\text{signal}}}{P_{\text{noise}}}$

* SNR_{dB} = $10 \log_{10} (\text{SNR})$

* Propagation time = $\frac{\text{distance}}{\text{speed} \rightarrow \text{light speed}}$

* Transmission time = $\frac{\text{message (bits)}}{\text{speed or bit rate}}$

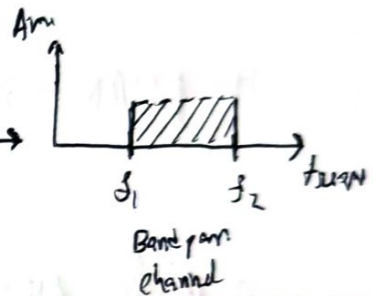
* Latency = Propagation time + Transmission time + Queuing time + Processing time.

* Nyquist $\rightarrow 2 \times \text{Bandwidth} \times \log_2 (\text{levels})$
 $\Rightarrow 2 \times \text{Bandwidth} \times \text{bits}$

[maximum bit rate] $\leftarrow$ Theoretical (noiseless)

* Shannon $\rightarrow \text{Bandwidth} \times \log_2 (1 + \text{SNR})$

[Noisy channel]
[noise level, SNR $\neq \infty$]
[avg. data rate]



Line Coding

Unipolar $\rightarrow$ NRZ $\rightarrow$ 0 $\rightarrow$ 0
1 $\rightarrow$ +ve

Lowest Bandwidth

long sequence of 0 & 1 (synchronization)

Polar $\rightarrow$ NRZ-L $\rightarrow$ 0 $\rightarrow$ +ve
1 $\rightarrow$ -ve

lack of synchronization

NRZ-I $\rightarrow$ 0 $\rightarrow$ no inversion
1 $\rightarrow$ inversion

inversion means opposite

MAN $\rightarrow$ 0 $\rightarrow$ $\neg$
1 $\rightarrow$ $\sqrt{}$

Highest Bandwidth

lack of synchronization

DMAN $\rightarrow$ 0 $\rightarrow$ inversion then transition
1 $\rightarrow$ no inversion

High Bandwidth

Bipolar AMI $\rightarrow$ 0 $\rightarrow$ 0
1 $\rightarrow$ +ve $\rightarrow$ -ve

Pseudoternary $\rightarrow$ 0 $\rightarrow$ +ve $\rightarrow$ -ve
1 $\rightarrow$ 0

[opposite of AMI]

MLT-3 $\rightarrow$ 0 $\rightarrow$ continue / no change
1 $\rightarrow$ +ve $\rightarrow$ 0 $\rightarrow$ -ve

Scrambling

Consecutive 8's zero error, last 1 error

B8ZS $\rightarrow$

-	0	0	0	-	+	0	+	-	
+	0	0	0	+	-	0	-	+	

HDB3 $\rightarrow$ zero error substitution rule

AMI rule $\rightarrow$ 1 positive to negative

Substitution $\rightarrow$ even or odd

Even $\rightarrow$ 000V

Odd $\rightarrow$ 000V

B means by the rule

V means violation (Rule 1)

* প্রতিটি point এর closest mid point এর Quantization code $N(M)$ serially লিখলে পাওয়া যায় - (2 5 7 6 2 1 2)

* এই Quantization code এর assigned binary N কোড (যা 010 101 111 110 010 001 010)

* এটিই ডাটাস্ট্রিম binary coded signal যা D-MAP দ্বারা ২৮

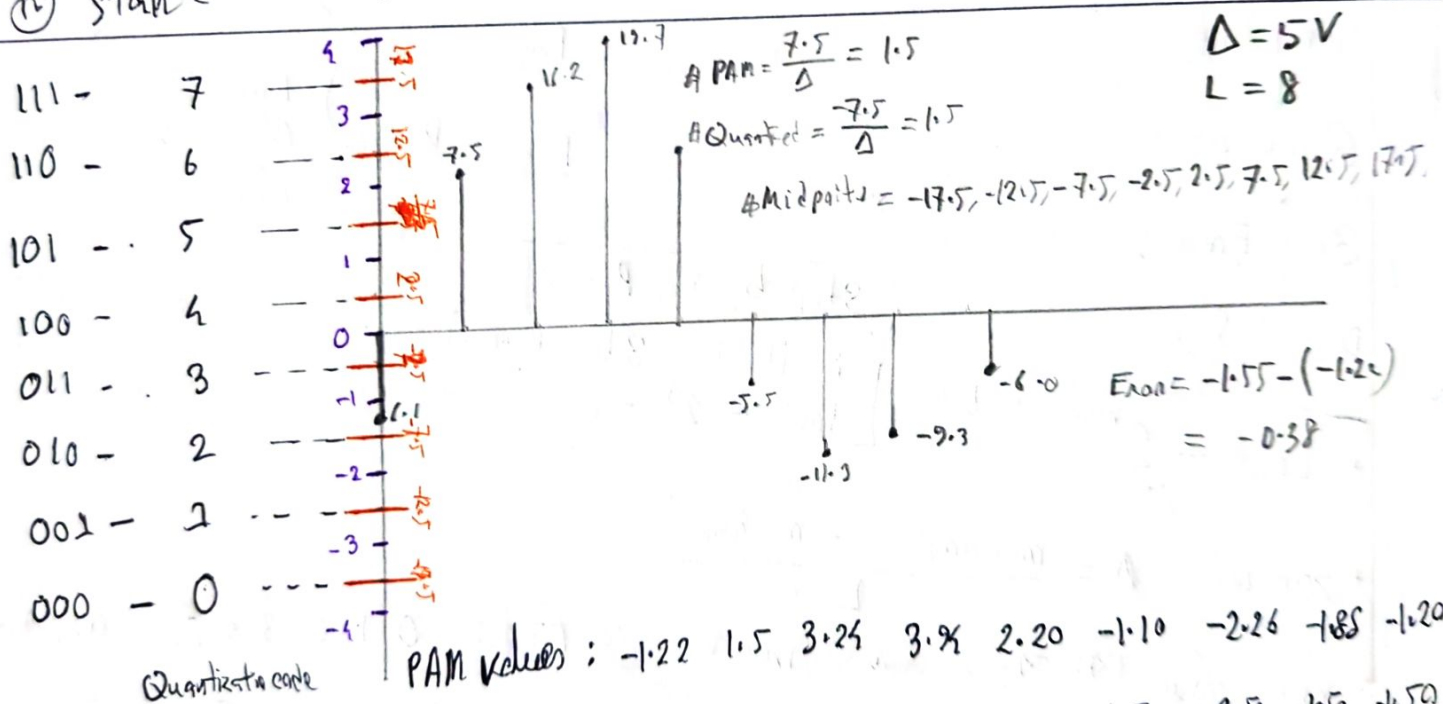
Staircase

① Question এর graph এর উপর আঁকলে Δ বাক্সে চিত্রিত হয়।

② Signal এর উপর লিখা লোক স্থাপন করলে ২৮ ~~০~~ ০।

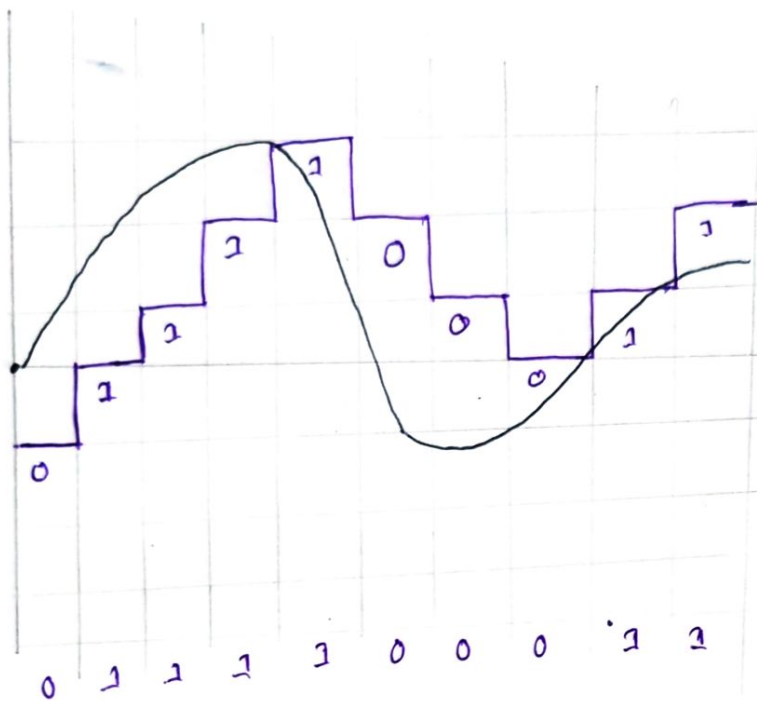
③ লোক signal উপর চিত্রিত করে ১ লোক লিখা যায় ২৮ ০।

④ Stair case আঁকলে ২৮ পাওয়া যায়।



Quantization code	PAM values	Quantized value	Normalized Error	Quantization Code	Encoded Word
111 - 7	-1.22	-1.50	-0.38	2	010
110 - 6	1.5	1.50	0	5	101
101 - 5	3.24	3.50	0.26	7	111
100 - 4	3.9	3.50	-0.44	7	111
011 - 3	2.20	2.50	0.30	6	110
010 - 2	-1.10	-1.50	-0.40	2	010
001 - 1	-2.26	-2.50	-0.24	1	001
000 - 0	-1.85	-1.50	0.35	2	010
	-1.20	-1.50	-0.30	2	010

Stair Case:



Baseline Wandering:

Loss of Synchronization: receiver cannot find data in the stream.

Ques: 



અનિ હોય તો અવરોધ 0 અથવા 1. આથી
average signal એ લાઇન શિફ્ટ અને અનિ data લેવા માટે

Possible Cases: NRZ, NRZ-L, AMI (comnegative 0), Pseudoternary (comnegative 1)

De Componente:

DE Component:
 signal constant zero. spectrum very low frequency (30) or more or demand
 zero. or 30% baseline wandering??

Possible causes: NRE, NRE-L, AMI (conjugative 0), Pseudotumor (conjugative 1)